

- **Problem Statement ID – SIH26090**
- **Problem Statement Title - AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans**
- **Theme - Heritage & Culture**
- **PS Category - Software**
- **Team ID -**
- **Team Name - RUBIXCUBE**



Problem - Traditional craft-based micro-entrepreneurs sales stay tied to physical channels such as middle-men or government fairs. Because of **gaps in digital literacy, language and e-commerce skills** they **struggle to transition to continuous digital markets**.



Proposed Solution - A **voice-first mobile app** where the artisan **photographs their product** then **answer the questions** about their product in their own language, the **app writes SEO-friendly product descriptions** and **lists it onto marketplaces**, B2B buyers and GeM.

How it Addresses the Problem:



- **Capture & Upload** : The app guides product placement and framing, helping artisans capture and upload the best possible photo from their phone.



- **AI Image Enhancer & Studio** : AI automatically removes cluttered backgrounds, corrects lighting, and enhances the photo to meet e-commerce standards.



- **Multilingual Auto-Cataloger** : Artisan describes the product through voice in their regional language, and AI creates SEO optimized descriptions in English and Hindi.



- **Dynamic Pricing Assistant** : A ML algorithm that analyzes the uploaded product image and description to suggest an optimal, competitive selling price based on current market trends and raw material costs.



- **List to Marketplace** : AI combines the enhanced image, product details, and price into a complete catalogue and prepares it for marketplace publishing.

Unique Selling Point



One Tap Listing
to e-market places

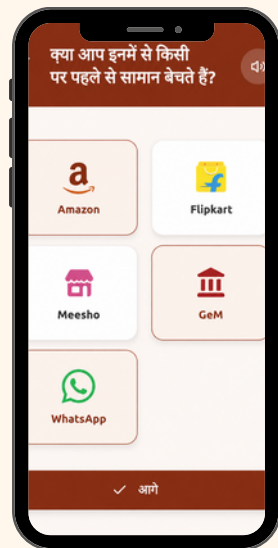


Self-Improving AI Agent
Your Business Manager



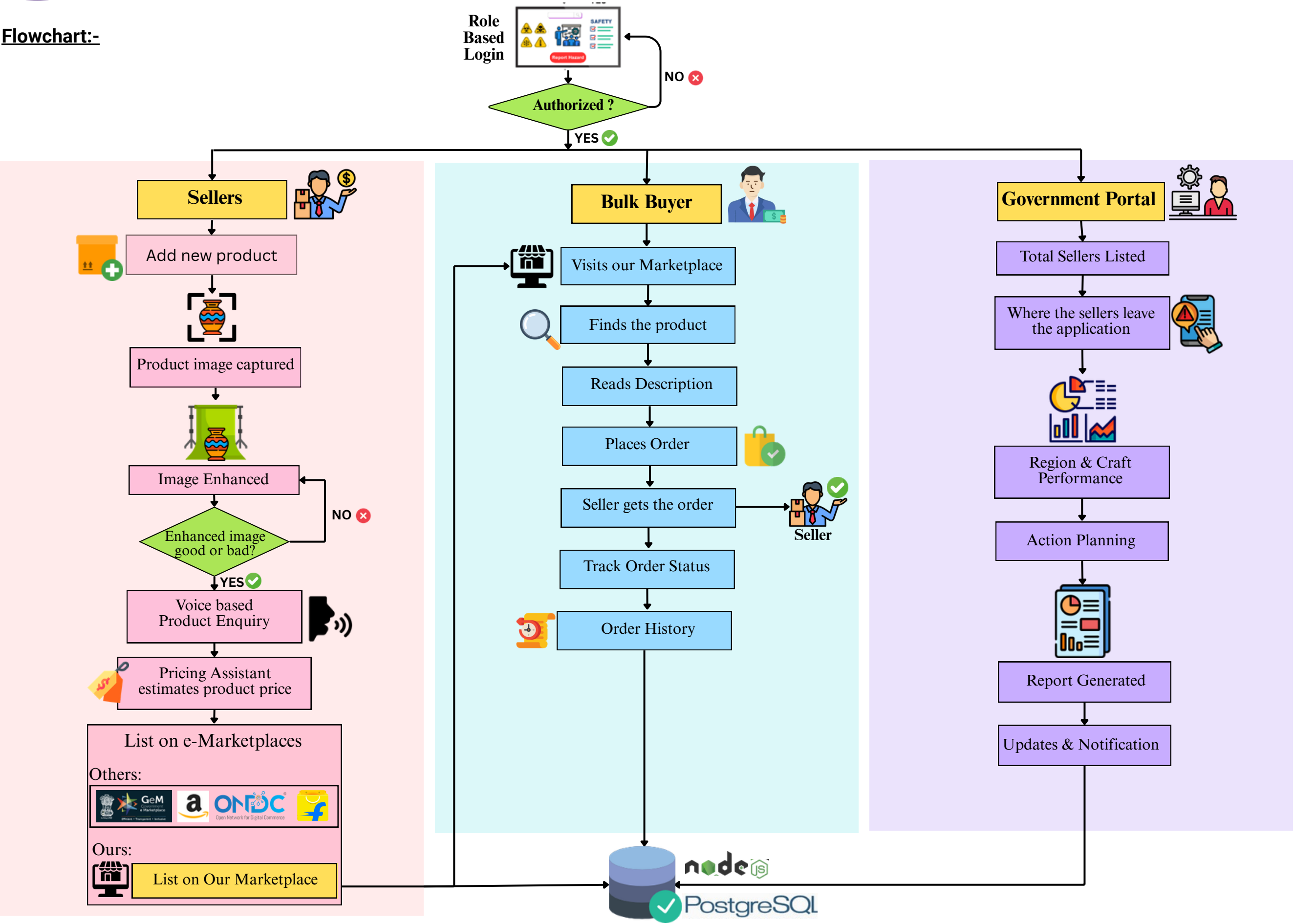
Voice - First Input
with minimal typing

Our App Interface



TECHNICAL APPROACH

Flowchart:-



Tech Stack

Frontend & UX

Backend & Data

AI & Product Intelligence

Voice, Marketplace & Cloud

sarvam

Our Video Explanation:
Solution Demo & Workflow
Explanation: [Here](#)
Complete Presentation: [Here](#)



FEASIBILITY

Technical Feasibility: **Bhashini & LLMs** for image enhancement, **multilingual-voice assistance** for catalogue management.

Economic Feasibility: AI-assisted cataloging reduces **manual digitization costs** and listing effort.

Operational Feasibility: **Voice-first workflow** and **camera guidance** simplify product cataloging.

Data & Privacy: Role-based access, **consent-driven data collection**, and secure storage protect artisan, buyer and program data

VIABILITY

Market Scope: **Marginalized artisans**, weavers and craft-based micro-entrepreneurs seeking digital market access.

Impact: Converts physical products into **structured digital catalogues**, enabling continuous **digital market access**.

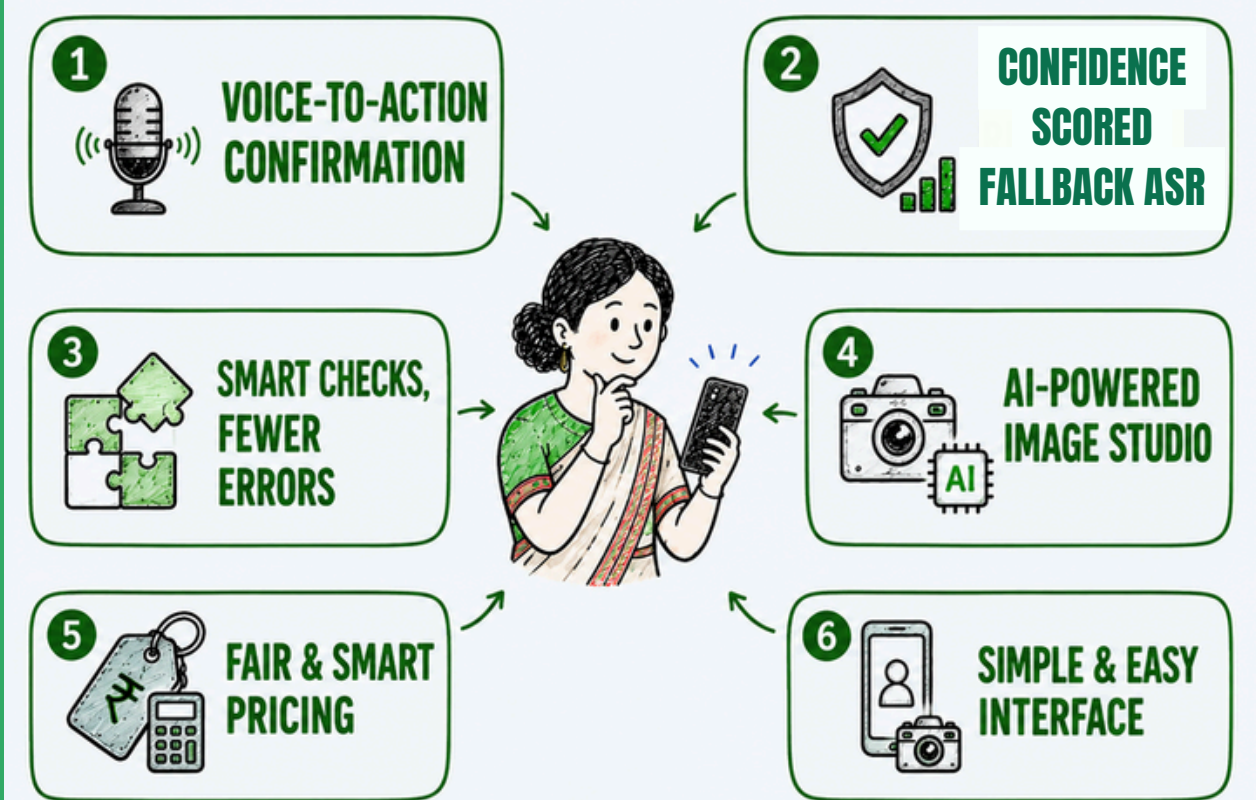
Scalability: The modular system **can easily extend to new crafts**, languages and marketplace channels.

Sustainability: AI-assisted cataloguing **reduces repeated work** while ensuring long-term **digital ecosystem integration**.

POTENTIAL CHALLENGES



PROPOSED SOLUTIONS



CHANNELS

- Government e-Marketplace
- B2B Buyers
- ONDC / Open Marketplaces
- Individual Marketplaces

REVENUE SOURCES

- Government Support
- White Label Licensing
- Marketplace

Impacts Created

ARTISANS

- Minimal-Typing Digital Inclusion (Voice & AI)
- Continuous Digital Market access (ONDC & GeM)
- Better earning potential



B2B BUYERS

- Smart Capacity Splitting for urgent bulk
- Production-Time Visibility
- Direct Artisan-to-Buyer Access

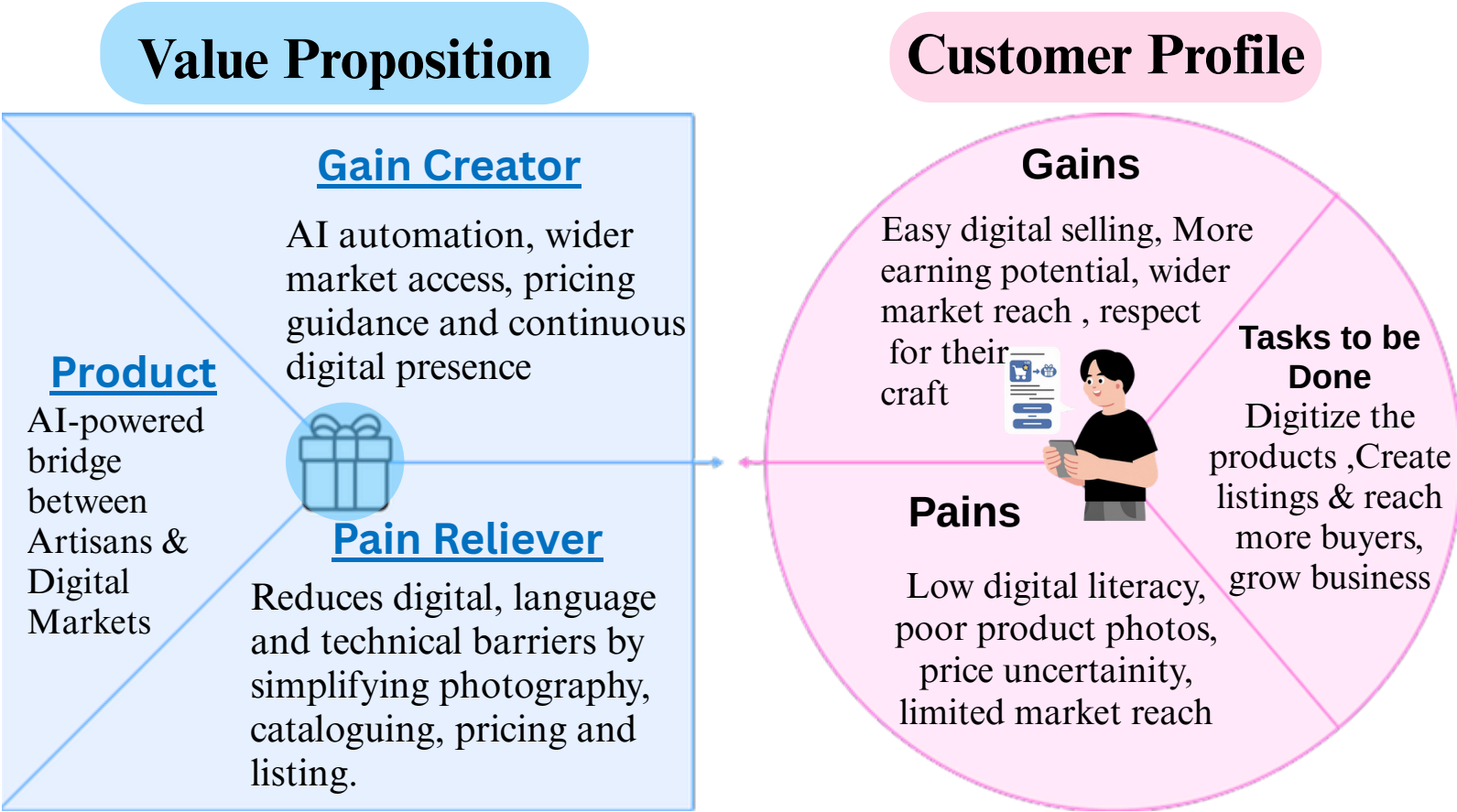


GOVERNMENT & AGENCIES

- Scalable Digital Onboarding
- Monitor Digital adoption & Market participation
- MoSJE Beneficiary Impact Tracking



IMPACT AND BENEFITS



FILLING THE DIGITALIZATION GAP: COMPLEMENTARY, NOT COMPETITIVE

Parameters	Our Solution	GeM MarketPlace	Destination Marketplaces
USP	AI-powered artisan digitization	B2G Procurement Portal	External B2B/B2C Marketplaces
Innovation	Voice-First Product Input	Complex Onboarding Layer	Manual Seller Onboarding
Seamless integration	One click export to multiple channels	No Public API – Excel Upload Only	Manual, Separate Listing Per Platform
Key features	Voice-to-Listing Pipeline, AI Image Studio, Dynamic Pricing	Government Procurement Portal	Standard Listing, No AI Assist

Benefits

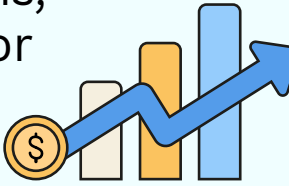
SOCIAL

Empowers marganalized artisans, promote digital inclusion & preserves traditional crafts, leading to self reliance and dignity.



ECONOMIC

Higher income potential for artisans, efficient sourcing for buyers & reduced operational costs through AI Automation



ENVIRONMENTAL

Reduces travel-related emissions through digital market access & consolidated shipments.

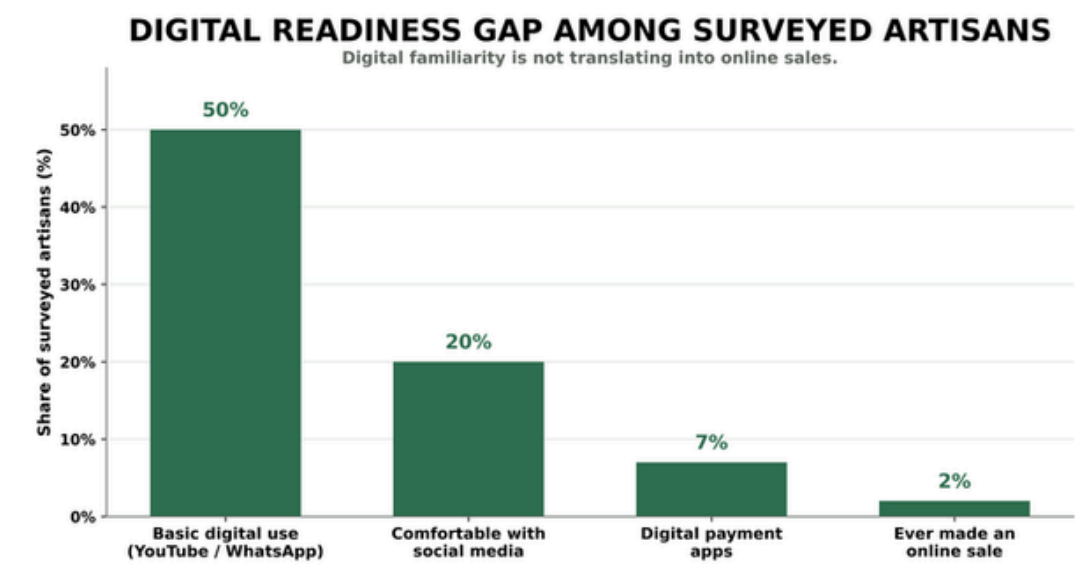
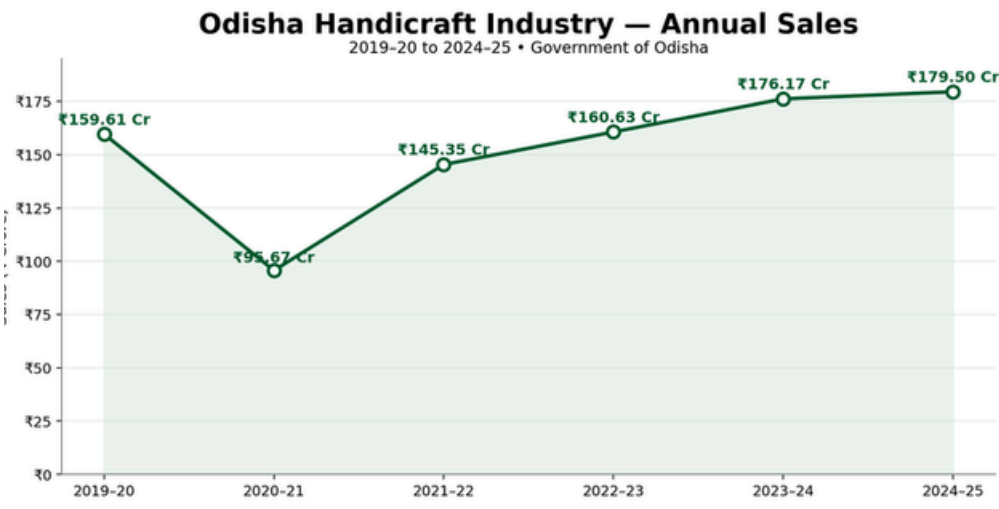
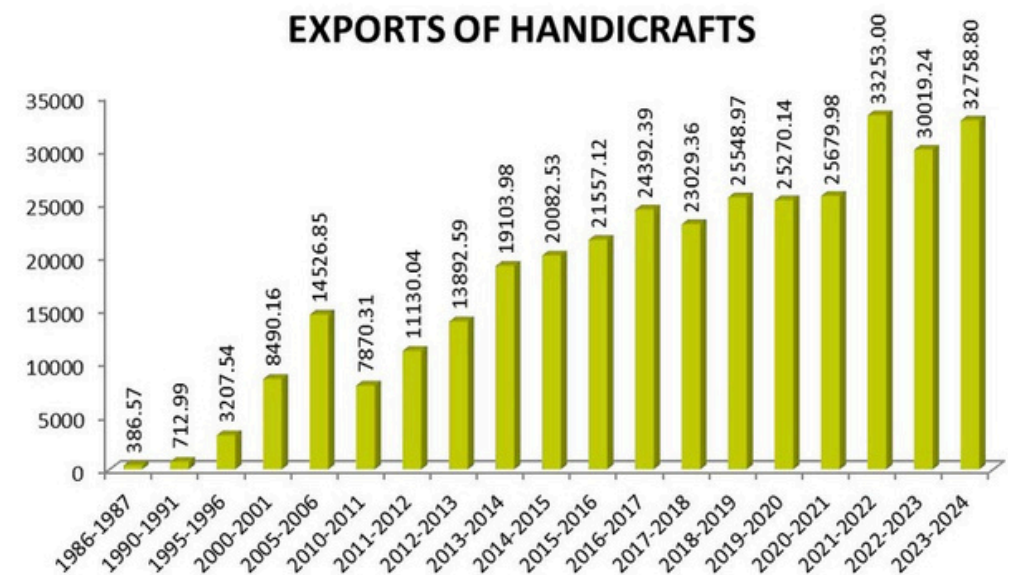


RESEARCH AND REFERENCES

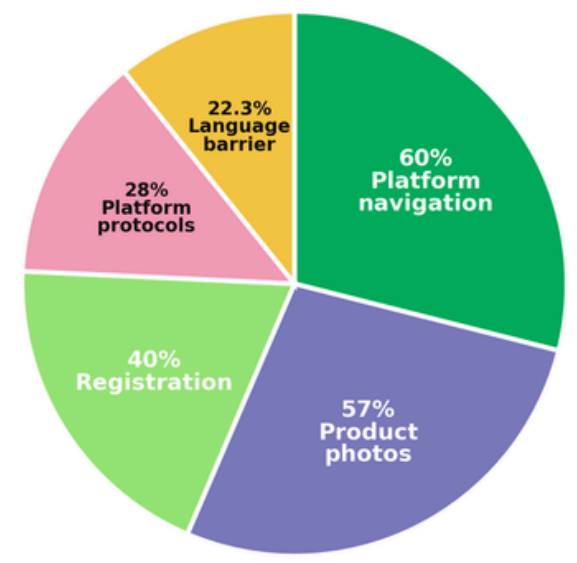


Research Finding

- 1. Voice-First Interfaces Improve Accessibility - [1]**
Voice and graphical interfaces improve accessibility for low-literacy users.
- 2. Significant E-Commerce Adoption Gap - [2]**
Few surveyed artisans sell online, highlighting low digital-commerce participation.
- 3. Large Digital-Commerce Opportunity - [3]**
India has 64.66 lakh handloom & handicraft artisans, offering strong digital-market potential.

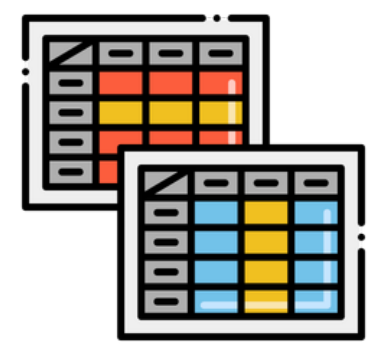


BARRIERS TO ONLINE SELLING
Key challenges faced by artisans in using digital platforms



Refrences

- [1] [Medhi et al. \(2007\) - Microsoft Research](#)
Research on voice and graphical interfaces for low-literacy and semi-literate users
- [2] [India Development Review \(IDR\)](#)
Artisan E-commerce Study Research on digital-commerce adoption.
- [3] [PIB - Indiahandmade](#)
Artisan population and handicraft sector statistics
- [4] [Catalyst Impact Accelerator](#)
Artisan Digital Readiness Study
- [5] [EPCH / Ministry of Textiles / PIB](#)



KEY TAKEAWAYS

- Digital access ≠ digital commerce
Basic digital use does not translate into online selling. [2][4]
- Product digitization remains a barrier
Photos, navigation, registration and listing create key challenges. [4]
- Market infrastructure already exists
IndiaHandmade and ONDC provide access; easier enablement is still needed. [3][6]

